42577 Introduction to Business Analytics course

Project statement

Welcome to this year's challenge!

The topic this year is *energy (and mobility)*. At a time when the world is facing unprecedented challenges of different kinds, including climate change, pandemics, social inequality, and degrading biodiversity, electric vehicles (EVs) present themselves as essential for reducing greenhouse gas emissions and combating climate change, as they produce zero tailpipe emissions and can be powered by renewable energy sources. Additionally, EVs offer long-term economic benefits by reducing reliance on fossil fuels, lowering fuel costs, and improving air quality, contributing to healthier communities.

In this project, we invite you to step in the shoes of an EV charging provider and use your best Business Analytics skills to help the company manage its operations and plan its future developments, thus providing a better service to its users and enhancing its business model. We want you to address the mandatory questions (below) but should also seek new questions, new data, and new insights.

You have access to a dataset that provides comprehensive insights into the usage patterns of electric vehicle (EV) charging stations in the city of Palo Alto, California. This dataset offers valuable information about the availability, utilization, and demand for EV charging infrastructure, allowing analysts to gain a deeper understanding of the local EV market and sustainable transportation practices. The data includes details on charging station locations, types of charging connectors available, charging session durations, energy consumption, and other relevant metrics. This information enables users to analyze trends, identify peak usage times, assess charging infrastructure performance, and develop data-driven solutions to enhance the efficiency and accessibility of EV charging services.

Project

The project has three components:

- *Prediction Challenge*: All groups must address the same problem.
- *Exploratory component*: Each group is invited to choose its own research questions and explore the data accordingly.
- *Report:* Each group should deliver one report in a paper format – use the IEEE double-column paper template (Manuscript Templates for Conference Proceedings | IEEE – or the one uploaded with this description on DTU learn). The maximum length is 6 pages. Besides this paper, you must also submit one (or multiple, if you prefer) well-structured and well-documented Jupyter notebook with all the code on which the report was based. The notebook must be self-explanatory and easy to follow and navigate. There should be a clear link between the sections/results in the paper and the corresponding source code in the

notebook. The paper should follow the 5-part outline shown below. The suggested length of each section is indicative and intended as a guideline.

Section 1: Introduction + motivation (0,5 pages)

Section 2: Data analysis and visualization (1 page)

Section 3: Prediction Challenge (2 Pages)

Section 4: Exploratory Component (2 Pages)

Section 5: Conclusions (0,5 Pages)

At the end of this document, you will find a list of practical information, which will include details on what is expected in each task and how these aspects contribute to the final grade. Section 2 and 3 can be further divided into methodology and result sub-sections.

- **Introduction to the data**

Each row in the dataset corresponds to an EV charging event by a user. Therefore it will have a user ID associated, a charging station (location), the particular plug used (a charging station can have multiple plugs), a start time when the EV is plugged in, an end time when the charging stops and the actual time when the EV is unplugged (these two can be very different due to "hoarders"), duration of the charging, energy consumption, etc. The full list of variables that you will have in this dataset is the following: Station Name, MAC Address, Org Name, Start Date, Start Time Zone, End Date, End Time Zone, Transaction Date (Pacific Time), Total Duration (hh:mm:ss), Charging Time (hh:mm:ss), Energy (kWh), GHG Savings (kg), Gasoline Savings (gallons), Port Type, Port Number, Plug Type, EVSE ID, Address 1, City, State/Province, Postal Code, Country, Latitude, Longitude, Currency, Fee, Ended By, Plug In Event Id, Driver Postal Code, User ID, County, System S/N, Model Number. The names of the variables should be more or less self-explanatory but don't hesitate to ask for clarification if you have any doubt about them.

The data is provided as a CSV file. Notice that the variables require some treatment in order to be usable (e.g. Dates, categorical, strings, different scales, IDs).

The *mandatory component* consists of two tasks:

1. Cluster the users based on their usage patterns - users with similar usage behaviors (e.g., charging more on the weekends, or more in daytime vs. nighttime) should be grouped together. The goal is to segment the user base, such that you can try to identify a target audience for the company's upcoming ad campaigns. (Important: note that the dataset provided contains charging events - you will have to transform that data into hourly energy use and hourly charger occupation for this analysis and also for Task 2). In the end, you should be able to provide recommendations to the company – you will not be evaluated on that recommendation, but on the appropriate usage of the techniques that you learned in class to achieve that recommendation. Make sure to clearly motivate your choices along the way!

2. Build a prediction model that, at the end of a day, allows one to predict what the energy consumption for each charging station will be over the next 24 hours with a 1-hour resolution – i.e. not the total energy consumption for the next day, but how the

time-series of the consumption (with 1h resolution) is expected to look like for the next day (e.g., given demand data until midnight of day 1, predict the consumption for all 1h intervals (6-7am, 7-8am, …, 11-12pm) in day 2). This information is crucial for your company to manage the energy grid properly. You can choose to use a single model that predicts consumption for all stations at once or multiple independent models. It is up to you to decide how best to formulate this problem as a machine learning problem. Whatever modeling approach you choose, your forecasts should have a lower error than a naïve baseline model that makes a prediction based on the historical average – e.g., use the time series of how an "average Monday" looks like (according to historical data in the training set) as the prediction for the next Monday in the testset. You are expected to implement this baseline approach and compare your prediction model to it. You should **not shuffle the data**. You should instead use the first 70% of the data to train your model, and the remaining data as a test set.

The *exploratory component* focuses on a strategic planning component. You are expected to leverage the data and modeling techniques covered in the course to provide recommendations to the company that help it improve on a more long-term horizon. Each group needs to address at least one new research question. We expect that you use at least one modeling technique (e.g. dimensionality reduction, clustering, classification, time series) in this component (i.e., you should not just use data handling and visualization to answer all your research questions). We expect you to formulate your own question, and follow the data sciences cycle. Feel free to make assumptions, but state them clearly. Some example topics that you may focus on are:

- Infrastructure planning. For example, where to build new charging stations or where to increase the capacity of the existing ones? What is the impact of land use on the charging demand (e.g., proximity to bus/metro station, shops, residential area vs. business district.)?
- Campaigns to increase revenue or improve resource utilization. This could be, for example, by providing discounts for charging at particular hours or at particular stations or by discouraging certain behaviors such as "hoarding" (how much potential revenue is the company losing due to hoarding?).

Note: The ordering of tasks we mention is **not** mandatory. In other words, if you prefer to start with the exploratory component and then go to the prediction challenge, this is also acceptable. You can mention that in the report (or invert Sections 2 and 3). Similarly, data analysis may be presented after the introduction (if relevant). However, please note that a simple descriptive analysis of the data is insufficient to complete the explanatory task. Make sure to go one step forward and try at least one of the techniques discussed in the course. Finally, in the exploratory component, you are expected to use at least one of the techniques listed in class (classification, regression, supervised, unsupervised, dimensionality reduction, …). Ideally, you should also benchmark multiple techniques (e.g., multiple regressors in the case of regression).

Evaluation

The evaluation of the paper+notebook(s) will be based on the following criteria:

- Clarity – clarity of the paper and the self-explanatory nature of the notebooks
- Appendices – We will mainly review the paper and only consult the notebook if needed. If space constraints prevent including certain plots or analyses, they may be added as appendices in the notebook. These must be clearly referenced in the paper and, where appropriate, briefly explained.
- Thoroughness - Each research question should be examined to a suitable depth. The explanatory component is expected to be developed with a level of detail comparable to the prediction challenge.
- Insightfulness - Don't stop at summarizing your findings — explain what kind of business insights or practical implications your results could provide.
- Technical aspects:
    - Data have been properly analyzed (data cleaning, data preparation, data pre-processing).
    - Which model has been used (only one model, multiple models, only linear models, or non-linear models)?
    - Is the model and approach appropriate?
    - Which performance metrics were used (how performances were evaluated)? Were they appropriate?
    - How was the approach benchmarked (how conclusions were drawn)?
- Honesty - While it's fine to use others' code (as a starting point), these shouldn't generally be the actual deliverable, **and** the appropriate ethical practice is to **always** reference the source of that code you used. Please add the relevant source material as a reference to the paper. **References do count towards the page limit**.

Rules

- Each group should consist of 4 students.
- The project submission shall be a zip file containing the paper PDF + all the notebooks. This zip file should contain the surnames of the group members (for example, for Pablo, Anders, Suarez, and Mila, it should be Pablo_Anders_Mila.zip).
- At the beginning of the paper, the abstract section should be used to **clearly clarify individual contributions**. In the event of doubts regarding individual contributions or the authenticity of the report, the teachers will call the group for an oral defense. **Individualized contribution section does count to the page limit**.

PLEASE INDICATE NAME, SURNAME, and STUDENT NUMBER IN THE REPORT. This should be indicated at the top of the article (author section). See the image below for an example

Deadline: November 30

42577 Introduction to Business Analytics

*Note: Sub-titles are not captured in Xplore and should not be used

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***Report Contribution*— All lecturers contributed equally to the organization of the course. Guido led the sections on unsupervised learning and dimensionality reduction; Ravi took the lead on supervised learning and classification; and Filipe played a key role in the advanced models component, particularly in neural networks. Other guy was a total free-rider**

I. Introduction (*Heading 1*)

This template, modified in MS Word 2007 and saved as a "Word 97-2003 Document" for the PC, provides authors with most of the formatting specifications needed for preparing electronic versions of their papers. All standard paper components have been specified for three reasons: (1) ease of use when formatting individual papers, (2) automatic

Keep your text and graphic files separate until after the text has been formatted and styled. Do not use hard tabs, and limit use of hard returns to only one return at the end of a paragraph. Do not add any kind of pagination anywhere in the paper. Do not number text heads-the template will do that for you.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

Figure 2. Example of report, with names, emails, student ID, and report contribution.